

file system created by unauthorized users or software attacks can be detected and removed. Using a tool of this type allows the administrator to locate the approximate time of attack since the modification will have taken place between two known timed events, the last and current execution of the integrity tool. Finally, integrity tools can be used to determine if a third party file has been modified or tampered with prior to use. Some manufacturers of Unix operating systems now publish MD5 digests of their systems. Using these digests, it is possible to determine that the file on your system is exactly as it should be. There was no degradation from misreading the installation media, deterioration of the disk system or intentional modification. If a manufacturer does not publish a list, then end users can create their own by installing an operating system on multiple systems from different media sources. The created digests of each system should agree.

Non-traditional Categories Of Protection And Their Failure

In the past, fencing systems were sold as a popular method of virus protection on PC platforms. A fencing system write protects parts of the disk using a hardware board that is added to the system bus. Since a virus cannot infect a file that is write protected using hardware, it appears to be a good method. The obvious drawback is that the user cannot write to the disk if it is write protected. The fencing system therefore had to create zones of protection so that the user could perform useful work. Viruses happily infected the unprotected zones. Fencing systems appear to have never been marketed for Unix systems.

Projection Of Future Problems

The problem of attack software written for and targeted against Unix systems will continue to grow, especially now that the Internet has gained popularity. Unix systems are the backbone of the world wide Internet. Viruses will become more prevalent because they provide all of the benefits of other forms of attack while having few drawbacks. Trans-platform viruses may become common as an effective attack. All of the methods cur-